Software Requirements Specification

AI-GRAFT V2

Software-Based Grafting Intelligence: Seedling Assessment, Graft-Quality Scoring, Early Failure Prediction and Healing Advisory Platform

Item	Details
Course	[Course name and code]
Instructor	[Instructor name]
Team name	[Team name]
Team members	[Member 1 – roll no.] [Member 2 – roll no.] [Member 3 – roll no.] [Member 4 – roll no.] [Member 5 – roll no.]
Project duration	[Confirm duration and submission date]
Version	2.0 — software-only revision
Date	8 October 2026
Initial crop	Tomato; extensible to cucumber and watermelon

Project proposition. AI-GRAFT V2 uses ordinary smartphone/webcam photographs and a printed reference marker to assess seedlings, guide a human grafting operator, score graft quality, monitor healing over time, estimate failure risk, and provide evidence-bounded healing advice. No robot, actuator, thermal camera, depth sensor, force sensor or automatic chamber control is required.

Document note. This document follows the supplied Plant Disease Detection SRS structure and limits functional requirements to five top-level items. Related capabilities are grouped under those five requirements.

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for AI-GRAFT V2, a software-centered agricultural AI prototype. It provides a common reference for planning, design, implementation, testing, demonstration and evaluation.

1.2 Document Conventions

Requirements are numbered FR-1 to FR-5, stated using “shall”, and assigned Must or Should priority. Must items define the minimum viable submission; Should items follow only after Must items work. Quality targets are evaluation goals, not claims of achieved performance.

1.3 Intended Audience and Reading Suggestions

This document is intended for developers, project coordinators, testers, AI/ML researchers, agronomy collaborators and evaluators. Read Section 2 for scope and constraints, Section 3 for testable requirements, then Sections 4–5 for behavior and implementation planning.

1.4 Product Scope

AI-GRAFT V2 is a local-first web application for tomato grafting research. It supports image capture/import, marker-based image calibration, morphology and readiness estimation, rootstock–scion comparison, human-facing cutting guidance, graft quality scoring, time-lapse healing monitoring, early failure-risk estimation, and bounded healing-condition advice. Human operators perform all physical grafting. Inputs, predictions, model versions and observed outcomes are recorded for validation.

The project does not claim novelty from cutting-point detection alone. Its proposed contribution is a software-only workflow combining calibrated image measurements, interpretable graft-quality scoring, uncertainty-aware abstention, temporal failure prediction and evidence-bounded advisory output.

2. Overall Description

2.1 Product Perspective

The product is a standalone local-first web application with a Python AI service and persistent database. It runs on a normal laptop/workstation and uses smartphone/webcam images. GPU is optional for training; core inference should not require continuous internet access.

Layer	Components	Purpose
User interface	Web dashboard, capture/import, graft records and reports	Operator workflow
Image processing	Quality checks, marker calibration, segmentation, keypoints and tracking	Consistent measurements
AI/ML	Readiness, pair ranking, GQI, temporal failure model	Decision support
Advisory	Rules and experiment-bounded recommendations	Advice without hardware control
Data management	Database, image storage, annotations and model registry	Traceability

2.2 Product Function

Five grouped functions are defined: (1) plant image intake and assessment; (2) graft planning and rootstock–scion decision support; (3) graft quality scoring and lifecycle records; (4) time-lapse healing and early failure prediction; and (5) healing advice, explanations and research reporting. Related capabilities are grouped so the SRS contains no more than five functional requirements.

2.3 User Classes and Characteristics

User class	Characteristics	Needs
Nursery operator	Performs grafting manually	Capture guidance, scores, cut guidance and alerts
Agronomist/researcher	Plant science and experimental knowledge	Outcome labels, explanations, comparisons and exports
ML developer	Builds/evaluates models	Dataset splits, metrics, versions and reproducibility
Administrator	Maintains app and data	Configuration, access control, backups and audit

2.4 Operating Environment

Client: current Chrome, Edge or Firefox on laptop/desktop; phone images may be uploaded. Image source: ordinary smartphone/webcam, preferably 1080p, plus a printed marker of known dimensions. Backend: local workstation with Python inference and persistent storage. Internet is optional except for downloading datasets or optional services.

2.5 Design and Implementation Constraints

The initial crop is tomato. Single-view RGB supports calibrated in-plane measurements only; likely out-of-plane geometry must be flagged. Cultivar, lighting, occlusion, graft clips and label availability affect performance. Biological success cannot be guaranteed. Healing advice is advisory and limited to documented or experimentally supported ranges. Dataset splits must be plant/batch/date grouped to reduce leakage.

3. Specific Requirements

Priority: Must = required for initial submission; Should = optional after Must items work. Exactly five top-level functional requirements are specified below.

3.1 Functional Requirements

ID / Priority	Requirement	Description and acceptance criteria
FR-1 / Must	Seedling intake, calibration and assessment	The system shall let a user create a batch, capture/import RGB images, associate crop/cultivar/plant IDs, check image quality, use a printed marker for scale, and estimate stem/hypocotyl geometry and graft-readiness. It shall flag unusable images or uncertain/out-of-plane measurements instead of returning unjustified confidence.
FR-2 / Must	Graft planning and rootstock–scion decision support	The system shall display candidate rootstock–scion pairs using entered variety information, measured morphology and available outcome history. It shall show cutting-point/angle guidance, supporting measurements and confidence. Recommendations are advisory; a human performs the graft.
FR-3 / Must	Graft quality scoring and lifecycle record	The system shall record a graft event and provide a documented Graft Quality Index (GQI) using observable features such as alignment, diameter match, visible cut/union condition and operator observations. It shall link pre/post images, overrides and eventual outcome to one plant record.
FR-4 / Must	Time-lapse healing and early failure-risk prediction	The system shall associate repeated dated photographs with a graft record, score visible healing progression and estimate failure risk over a defined horizon. It shall show uncertainty and latest observation time. Estimates shall be evaluated on held-out plants/batches; insufficient evidence shall be labelled experimental or unavailable.
FR-5 / Must	Healing advisory, explanations and research reporting	The system shall provide bounded, non-actuating healing advice from entered conditions, optional environment logs and documented evidence. It shall explain key factors where supported, allow authorized users to correct labels/record outcomes, and export records, predictions, model versions and evaluation summaries. It shall not control a chamber or guarantee survival.

3.2 Software Interfaces

The interface shall communicate with a Python inference service through a documented local REST API or equivalent. Use persistent metadata storage (SQLite for single-user prototype or PostgreSQL for multi-user use) and file storage for images. OpenCV, PyTorch/TensorFlow and scikit-learn/XGBoost are candidate implementation libraries, not mandatory dependencies.

3.3 Communications Interfaces

Local browser-to-backend communication shall use localhost or a trusted local network. If remote access is enabled, HTTPS and authentication shall be used. Images and predictions shall be linked through stable IDs and timestamps. Environment logs may be imported as CSV/JSON; live IoT or robot control is not required.

3.4 Non-Functional Requirements

ID	Quality attribute	Minimum target / verification
NFR-1	Response time	Aim for inference within 5 seconds for a standard 1080p image on the target workstation; report median and P95 over at least 100 images.
NFR-2	Model quality	Report segmentation/keypoint metrics and morphology MAE/R^2 against expert labels. Report failure AUROC/AUPRC, calibration and lead time only where sufficient outcomes exist.
NFR-3	Reliability and abstention	Invalid, blurred, occluded or unsupported images shall return a warning or abstain; low-confidence output shall not be shown as confirmed outcome.
NFR-4	Data integrity	Link plant IDs, batches, annotations, predictions, outcomes and model versions; use grouped data splits to reduce leakage.
NFR-5	Usability, privacy and maintainability	Common workflows shall be usable without command-line input; protect local data and document modular code/models.

4. System Features

4.1 Seedling Intake and Assessment

Stimulus: an operator creates a batch and captures/uploads an image. Response: the system checks image quality, identifies plant regions/keypoints, applies marker-based scale calibration when possible, and displays measurements/readiness with confidence. Missing markers or poor images trigger recapture or an uncalibrated warning.

4.2 Graft Planning and Quality Scoring

Stimulus: the operator selects a candidate pair and records a graft. Response: the system displays geometry-based cutting guidance and supporting evidence. After grafting, the operator submits an image and optional observations; the system produces a GQI and allows the record to be updated when the survival outcome is known.

4.3 Healing Monitoring and Failure Prediction

Stimulus: a user adds a dated follow-up photograph. Response: the system updates the healing timeline, reports visible stage indicators and estimated failure risk over a configured horizon, and displays uncertainty. If training/follow-up evidence is insufficient, the estimate is marked experimental or unavailable.

4.4 Healing Advisory and Research Dashboard

Stimulus: a user enters current healing conditions or imports a CSV/JSON environment log. Response: the system presents recommendations only within documented evidence bounds, explains limitations, records advice and permits export of records/outcomes/predictions/model versions. No advice is sent to an actuator.

5. Project Plan

5.1 Schedule Basis

The following is a proposed four-week prototype plan and should be aligned with the instructor's actual deadline. Longitudinal data collection must begin early because survival labels arrive after grafting.

5.2 Work Modules

Module	Work	Suggested lead
A. Data and imaging	Collect images/outcomes, define protocol, marker calibration, splits and annotation guide.	[Name]
B. Vision and morphology	Image quality, segmentation/keypoints, morphology and readiness baseline.	[Name]
C. GQI and prediction	Quality score, follow-up timeline, baseline failure model and evaluation.	[Name]
D. Application	Web UI, API, plant records, dashboard and exports.	[Name]
E. Testing/report	Integration tests, evaluation, literature review, README, report and slides.	[Name]

5.3 Gantt-Style Plan

Task	Week 1	Week 2	Week 3	Week 4
Setup, literature and data protocol	●	●		
Image collection, marker and annotation	●	●	●	
Baseline image assessment	●	●	●	
GQI and record workflow		●	●	
Temporal failure baseline		●	●	●
Dashboard/API integration		●	●	●
Testing and error analysis			●	●
Report and final demo			●	●

5.4 Milestones

Milestone	Expected outcome
End Week 1	Repository, data protocol, plant IDs and first annotated images ready.
End Week 2	Working image intake/calibration and baseline morphology/segmentation.
End Week 3	GQI and dashboard integrated; follow-up records collected; baseline prediction evaluated if labels permit.
End Week 4	End-to-end demo, measured results, limitations, report and slides complete.

5.5 Deliverables

Source-code repository and README; environment/requirements file; dataset documentation and annotation guide; model weights or reproducible training instructions; working demo; sample plant/graft records; evaluation tables/error analysis; SRS, final report and presentation.

5.6 Risks

Risk	Mitigation
Too few survival/failure labels	Start collection early; report pilot results and avoid overclaiming.
Camera/lighting variation	Use marker and capture guide; test on held-out devices/batches.
Union hidden by clip/leaves	Support recapture and abstention; record visibility limitations.
Data leakage	Split by plant/batch/date; preserve untouched test set.
Non-causal healing advice	Keep advisory; restrict to documented/experimentally varied conditions.
Schedule slips	Complete Must requirements first; postpone advanced ML.

6. What Is Not Included

The following are outside AI-GRAFT V2 scope: robot, motor, cutter, gripper or autonomous physical control; depth, thermal, hyperspectral or force/tactile sensors; automatic healing-chamber, humidity, temperature, lighting or irrigation control; guaranteed compatibility/survival/yield; broad cross-crop claims without separate evaluation; production-grade mobile/cloud deployment; and advanced future work such as reinforcement learning, digital-twin robotics, federated learning and cross-crop foundation models. These may be considered after validating the software-only prototype.

6.1 Team Members and Division of Work

Work area	Assigned member(s)
Vision / ML model development	[Name(s)]
Backend, database and API	[Name(s)]
Frontend and UI/UX	[Name(s)]
Data collection, annotation and agronomy validation	[Name(s)]
Testing, literature review and report	[Name(s)]

Appendix A. Decisions to Confirm

Decision	Why it matters	Status
Final crop/cultivar and pairs	Determines data collection and compatibility claims	To confirm
Availability of real graft outcomes	Required to validate GQI/failure prediction	To confirm
Capture protocol and marker design	Measurement reproducibility	To confirm
Model baseline	Choose lightweight vision and temporal/tabular baselines	To confirm
Web stack and local database	Determines implementation effort	To confirm
Team roles and deadline	Execution plan	To confirm

Reference basis

The technical basis includes research on tomato grafting-point estimation with YOLO-family models; ML-based tomato seedling growth regulation; reviews of vegetable grafting robotics and grafting success factors; and methods for fiducial-marker calibration, uncertainty calibration and survival analysis. Verify complete bibliographic records and conduct a focused literature/patent search before claiming novelty.

End of AI-GRAFT V2 Software Requirements Specification